Mechanism-Directed Drug Repurposing in BUB1B-Related Mosaic Variegated Aneuploidy

A set-point rebalancing hypothesis, and the case against the standard aneuploidy-targeting strategies

Nandini Gantayat

25 October 2026

Rare Disease, Real Kid: The MVA Hackathon 2026
Sage Bionetworks, MVA Society, Hugging Face, BEACON

Contents

0. Status of this document **2**

1. Executive summary **3**

2. Case and causal variant **4**
2.1 Clinical features 4
2.2 The variants 4

3. Molecular mechanism **6**
3.1 BubR1 is a pseudokinase 6
3.2 Two distinct safeguards, not one 6
3.3 The evidence this applies to MVA specifically 6

4. The proposal: set-point rebalancing **8**
4.1 Statement 8
4.2 The pharmacology is predicted to be non-monotonic 8
4.3 What would falsify it 8

5. Why the obvious alternatives are worse **9**
5.1 Increase BubR1 abundance 9
5.2 Exploit aneuploidy-associated stress 9
5.3 Enhance clearance of aneuploid cells 10

6. Supporting analyses **11**
6.1 Value-of-information for next-experiment selection 11
Mechanistic decision problem 11
Diagnostic decision problem 12
6.2 Mosaic aneuploidy burden from the call set 12

7. Clinical constraints **14**
7.1 Immediately actionable 14
7.2 Paediatric pharmacology, specific to this patient 14
7.3 Standard of care 14

8. Limitations **15**

9. Reproducibility **16**

10. References **16**
Acknowledgements 17

0. Status of this document

This is an analysis of the Track 2 question prepared during the *Rare Disease, Real Kid: MVA Hackathon 2026*. It was **not entered as an official Track 2 submission**; it is published in the accompanying repository as a public record of the approach, under the CC BY 4.0 terms the hackathon applies to participant outputs.

The Track 1 variant-prediction submission associated with this work scored 100 rank points and F-max 1.000, full match at rank 1.

Nothing here is medical advice. Every candidate discussed is a hypothesis for further investigation, not evidence that a medicine works.

AI assistance disclosure. Anthropic API, Claude, commercial terms, no training on customer content.

Data handling. No patient sequence data or genotype-scale derivative appears in this document or in the accompanying repository. Reported findings are limited to the named candidate variants and to chromosome-level aggregate statistics, which the challenge terms permit.

1. Executive summary

The causal genotype is compound heterozygous *BUB1B*: p.Leu737Ter, a nonsense allele predicted to undergo nonsense-mediated decay, in trans with p.Asn1002Lys, a missense allele in the C-terminal pseudokinase domain.

The central argument of this report is that **the standard therapeutic strategies for aneuploidy are pointed the wrong way in this patient**, and that a different intervention class follows from reading the lesion correctly.

Three claims, in order of importance:

First, the lesion is a degraded scaffolding interface, not a broken enzyme. BubR1 is a pseudokinase. Its C-terminal domain is catalytically inactive and functions by recruiting the PP2A-B56 phosphatase to the kinetochore. The defect is therefore graded rather than binary, which matters because graded defects in a balance are shiftable.

Second, the tractable intervention is rebalancing rather than killing. PP2A-B56 opposes Aurora B kinase in the kinetochore error-correction pathway. When BubR1 cannot recruit PP2A-B56, Aurora B runs unopposed and destabilises attachments that should have been stabilised. Published work in MVA patient-derived cells shows this congression defect is reverted by Aurora B inhibition. The proposal is partial, low-dose restoration of that set point, with a predicted non-monotonic (U-shaped) dose-response that makes the hypothesis falsifiable.

Third, the aneuploidy-exploitation strategies imported from oncology assume a host-tumour asymmetry that is substantially collapsed here. Those strategies work because the tumour is aneuploid and the host is not. In a constitutional chromosomal instability syndrome, every tissue carries the same instability and the same proteotoxic burden. Proposals in that class require the selectivity window to be measured in this patient before they can be entertained.

One finding is immediately actionable and independent of all of the above: because this is a chromosomal instability syndrome, standard genotoxic chemotherapy and radiotherapy for the rhabdomyosarcoma carry elevated secondary-malignancy risk.

2. Case and causal variant

2.1 Clinical features

Feature	HPO ID
Rhabdomyosarcoma	HP:0002859
Nephrocalcinosis	HP:0000121
Short stature	HP:0004322
Failure to thrive	HP:0001508
Skeletal muscle atrophy	HP:0003202
Premature birth (32 weeks)	HP:0001622
Small for gestational age (~1 kg)	HP:0001518
Parental recurrent spontaneous abortion	HP:0200067

Two aspects drove the interpretation. **Cancer predisposition with growth restriction** is characteristic of chromosomal instability syndromes. **Parental recurrent pregnancy loss** was treated as primary inheritance evidence rather than background: under an autosomal recessive model with two healthy carrier parents, a quarter of conceptions are biallelic, and in a chromosome-segregation disorder the severe end of that distribution is expected to be embryonic lethal.

Noted discrepancy. Microcephaly is not reported here, although it is a prominent feature of *BUB1B*-associated MVA as originally described. This was treated as a reason to carry all three MVA genes (*BUB1B*, *CEP57*, *TRIP13*) through the analysis rather than restricting a priori. The genotype evidence did not support redirection; the discrepancy is recorded as unresolved.

2.2 The variants

	Variant (GRCh38)	HGVSc	HGVSp	Consequence	gnomAD AF
Allele 1	chr15:40209702 T>G	c.2210T>G	p.Leu737Ter	stop gained, exon 17/23	1.0e-4
Allele 2	chr15:40220612 T>G	c.3006T>G	p.Asn1002Lys	missense, exon 23/23	absent

Allele 1 introduces a premature termination codon well upstream of the final exon-exon junction, predicted to trigger nonsense-mediated decay and give effectively null output from that copy. Allele 2 substitutes a neutral polar asparagine for a positively charged lysine in the C-terminal pseudokinase domain; SIFT calls it deleterious, PolyPhen-2 probably damaging, and it is absent from gnomAD.

The null-plus-hypomorph configuration is consistent with the MVA literature, where complete biallelic loss of *BUB1B* is not observed in liveborn individuals, and with the parental reproductive history.

Phase is inferred, not observed. The alleles are 10,911 bp apart, beyond the span of a sequencing fragment, so no read-backed phasing links them. Trans configuration is inferred from the recessive disease model and the family history. Parental sequencing would establish it directly.

3. Molecular mechanism

3.1 BubR1 is a pseudokinase

The C-terminal domain of BubR1 adopts a kinase fold but has no catalytic activity. It functions as a scaffold: it recruits the PP2A-B56 phosphatase to the kinetochore. A damaging missense in this domain therefore degrades a protein-protein interaction surface rather than an active site.

This reframing has two consequences. The defect is **quantitative and graded**, so there is a dose of residual function rather than an on/off state. And the pathology is **loss of function**, for which inhibitor pharmacology — the thing small molecules are good at — is structurally mismatched.

3.2 Two distinct safeguards, not one

BubR1 is conventionally described as a spindle assembly checkpoint protein. That description is incomplete, and the incompleteness is where the therapeutic opportunity sits. Two separate mechanisms protect mitosis:

The spindle assembly checkpoint delays anaphase onset until every kinetochore is attached, via the mitotic checkpoint complex inhibiting APC/C-Cdc20. This is the “wait” signal.

The error-correction pathway is distinct. Aurora B kinase phosphorylates outer-kinetochore substrates to destabilise attachments lacking tension — syntelic and merotelic attachments — releasing them so the chromosome can re-attach correctly. Opposing Aurora B is PP2A-B56, recruited by BubR1, which dephosphorylates those substrates and stabilises attachments once correct.

Error correction is therefore a **kinase-phosphatase balance**, and BubR1 sits on the phosphatase side of it. The balance, not the scaffold, is the druggable object.

3.3 The evidence this applies to MVA specifically

Three results converge.

Work from the Kops and Medema groups identified BUBR1 as the recruiter of B56-PP2A, showed the complex opposes Aurora B, and demonstrated that failure of BUBR1 to recruit B56-PP2A contributes to the chromosome congression defects observed in cells derived from MVA patients — and that loss of the complex can be reverted by inhibiting Aurora B (reference 2).

A 2020 *Cell Reports* study localised the function: mutation or removal of the BUBR1 pseudokinase domain decreases PP2A-B56 recruitment to the outer kinetochore, attenuating checkpoint silencing and producing chromosome alignment errors as a result of imbalance in Aurora B activity. **p.Asn1002Lys lies in that domain.**

Reviews of kinetochore kinase-phosphatase cross-talk establish the general principle: removing kinetochore PP2A-B56 prevents formation of initial kinetochore-microtubule attachments because Aurora B activity is elevated.

The chain is therefore:

> pseudokinase-domain missense → reduced PP2A-B56 recruitment → unopposed Aurora B at the kinetochore → attachments destabilised when they should be stabilised → congression failure and missegregation → mosaic aneuploidy

and the published rescue for the penultimate step is pharmacological Aurora B inhibition.

4. The proposal: set-point rebalancing

4.1 Statement

The lesion has tipped a kinase-phosphatase set point. The phosphatase arm cannot readily be restored — there is no general pharmacology for increasing the abundance of a protein. The kinase arm can be partially inhibited, and approved and clinical-stage Aurora B inhibitors exist.

The therapeutic goal is to move the set point back toward equilibrium so that the residual BubR1 scaffold is sufficient for correct attachment. This is **preventive and chronic rather than cytotoxic and acute**: it acts upstream, on the generation of aneuploidy, rather than downstream on aneuploid cells.

4.2 The pharmacology is predicted to be non-monotonic

This is the central and most falsifiable prediction, and it is what distinguishes the hypothesis from the known cytotoxic use of the same drug class.

Aurora B inhibitors at oncology doses are cytotoxic and themselves cause chromosome missegregation; they are used experimentally as chromosomal instability inducers. The regime proposed here is different: **low-dose partial inhibition to restore a set point**.

The predicted dose-response is therefore **U-shaped** in missegregation rate:

- Too little inhibition: the Aurora B / PP2A-B56 imbalance persists, attachments remain destabilised, missegregation continues.
- Optimal partial inhibition: Aurora B activity falls to a level the residual PP2A-B56 recruitment can balance, and attachment stability is restored.
- Too much inhibition: error correction itself fails, syntelic and merotelic attachments are no longer released, and missegregation rises again — now by the opposite mechanism.

Locating that minimum is the decisive experiment. The existence of a minimum is what makes the hypothesis refutable: a monotonic dose-response would render the proposal indistinguishable from the already-known cytotoxic behaviour of the class and would falsify the set-point framing.

4.3 What would falsify it

- **Normal PP2A-B56 binding by the variant protein.** If p.Asn1002Lys does not reduce PP2A-B56 recruitment, the entire mechanistic chain collapses.
- **A monotonic dose-response** in patient-derived cells, as above.
- **Rescue failing to transfer to this allele.** The published rescue was shown in MVA patient lines carrying other mutations.

5. Why the obvious alternatives are worse

5.1 Increase BubR1 abundance

The mechanistic rationale is the strongest available for any strategy here. Baker et al. (2004) showed *BubR1* hypomorphic mice have high aneuploidy and age prematurely — the mouse model of MVA. The reciprocal experiment is the informative one: Baker et al. (2013, *Nature Cell Biology*) showed sustained BubR1 overexpression reduces chromosome missegregation, reduces aneuploidy, reduces tumorigenesis and extends lifespan. Follow-up work (eLife 2016) dissected which domains carry that benefit. The dose-response is clean and monotonic in the helpful direction.

The tractability is the weakest. There is no general pharmacology for "make more of a protein." The sub-routes each have defects:

- *Stabilise the variant protein.* Requires knowing its degradation route, which is unestablished for this variant.
- *Upregulate transcription.* The null allele contributes nothing, so only the missense allele can be upregulated — which helps only if that allele is hypomorphic rather than dominant-negative. This is currently unknown and is the pivotal open question for this route.
- *Nonsense read-through.* Aminoglycosides and related agents promote ribosomal read-through of premature termination codons. Efficiency is low and context-dependent, and chronic exposure is oto- and nephrotoxic. The renal toxicity is a serious contraindication in a child with nephrocalcinosis from birth.

This is the direction the biology endorses, with the honest statement that the tools do not exist.

5.2 Exploit aneuploidy-associated stress

The oncology literature here is substantial. Aneuploid cells carry unbalanced gene dosage, excess orphan subunits of protein complexes, and chronic proteotoxic burden. They show impaired HSF1 and HSP90 function, and synthetic-lethality screens identify elevated dependence on autophagy, the unfolded protein response and DNA repair. The implied drug classes — proteasome inhibitors, HSP90 inhibitors, autophagy inhibitors — are approved or in trials and therefore qualify as repurposing candidates.

Three objections, each sufficient on its own.

The host shares the vulnerability. This strategy works in sporadic cancer because the tumour is aneuploid and the host is not; that asymmetry *is* the therapeutic window. In a constitutional chromosomal instability syndrome, every tissue carries the same instability and the same proteotoxic burden. The window is substantially collapsed. Any proposal in this class that does not address this has

missed the defining feature of the case.

HSP90 inhibition may be actively counterproductive. Chen et al. (2012, *Nature*) showed that proteotoxic stress from transient Hsp90 inhibition is a potent inducer of chromosome instability and karyotype diversity. In a patient whose underlying pathology is chromosome instability, a drug that increases instability in surviving cells is a plausible accelerant of the next malignancy. Yeast evidence does not transfer automatically, but the direction of effect is wrong and the burden of proof sits with anyone proposing the class.

Selectivity is assumed, not measured. Whether this child's tumour cells are more proteotoxically stressed than his constitutional cells is an open empirical question. The honest form of this strategy is not a drug proposal but a measurement: quantify the differential stress before entertaining the class.

5.3 Enhance clearance of aneuploid cells

Aneuploid cell populations can become more euploid over time, and this is well documented in preimplantation embryos. Two mechanism families are described.

Selection, which dominates. Aneuploid cells proliferate more slowly, undergo apoptosis preferentially, and in embryos are preferentially allocated to trophectoderm rather than inner cell mass, diluting out of the fetal lineage. Mouse work shows mosaic embryos can be rescued when diploid cells are present at sufficient frequency. The embryo does not repair cells; it loses them.

True chromosomal rescue, which is rare. Trisomy rescue by extrusion of the extra chromosome, monosomy rescue by duplication of the remaining one — documented, including spontaneously during iPSC reprogramming of trisomy lines. But stochastic, and it generates uniparental disomy in roughly a third of cases, with attendant imprinting and recessive-unmasking risk.

Neither yields a therapy here, for three reasons.

It is overwhelmingly an **embryonic** phenomenon, operating in a rapidly dividing population with large redundancy. This patient is postnatal.

In MVA the checkpoint is defective **continuously**, so clearance competes against an ongoing source term. Faster clearance does not reach equilibrium at a euploid state.

Most importantly, **clearance is plausibly part of the pathology rather than the cure.** The BubR1 microcephaly literature reports massive cell death reducing cortical cell number during neurogenesis. In a child with severe intrauterine growth restriction, short stature and failure to thrive, aneuploid-cell apoptosis is already occurring and is a candidate driver of the tissue depletion. Enhancing selective apoptosis would remove cells he cannot afford to lose.

This argument is stated explicitly because "boost p53-mediated clearance of aneuploid cells" is a superficially attractive proposal that a reviewer may reasonably expect to see addressed.

6. Supporting analyses

Two analyses were implemented to support the argument. Both are included in the repository.

6.1 Value-of-information for next-experiment selection

Variant prioritisation pipelines return a ranked list. They do not say whether the ranking is adequately supported, nor what would resolve the remaining ambiguity. In an N-of-1 setting the operationally important question is not which variant to believe but which test to order next.

Prioritisation output is treated as a posterior over hypotheses, and each candidate experiment scored by expected entropy reduction:

$$\mathrm{VoI}(e) = H(C) - \sum_{r} P(r)\, H(C \mid e = r)$$

in bits, with $P(r) = \sum_h P(h)P(r \mid h)$ and $P(h \mid e = r)$ by Bayes. VoI is reported alongside cost and turnaround, since the clinically relevant ranking is information per unit of cost and delay.

All priors and likelihoods are explicit elicited subjective estimates, stated in the implementation so each can be challenged and substituted. They are not fitted and are not claimed to be calibrated.

Mechanistic decision problem

The diagnostic question closes once the genotype is confirmed. The mechanistic question — how much residual scaffold function p.Asn1002Lys retains — does not, and it is the assumption every therapeutic hypothesis above rests on.

Prior: M1 severe hypomorph (<20% residual) 0.40; M2 moderate (20-60%) 0.40; M3 mild 0.20. Prior entropy 1.522 bits.

Experiment	VoI (bits)	Fraction resolved	Days	Cost	Bits/cost
PP2A-B56 binding (co-IP)	0.546	0.36	14	3.00	0.18
Mitotic timing (live-cell)	0.376	0.25	21	4.00	0.09
In-silico stability	0.250	0.16	0.1	0.01	**25.04**
Aneuploidy rate	0.172	0.11	10	2.00	0.09

Two conclusions. **PP2A-B56 co-immunoprecipitation carries the highest absolute information** and is the correct next wet-lab commitment; it also directly tests the falsification condition in Section 4.3. **Structure-based stability prediction dominates on cost-normalised value by two orders of magnitude** and is the correct immediate action despite yielding less absolute information.

Decision boundaries for M1 via PP2A-B56 binding: low binding $\rightarrow$ posterior 0.79; partial $\rightarrow$ 0.13; normal $\rightarrow$ 0.06.

Diagnostic decision problem

Computed from the pre-submission evidence state and deliberately **not** conditioned on scored feedback, because the quantity of interest is what the pipeline recommends when no answer key exists. Prior entropy 1.681 bits over four configurations.

Experiment	VoI (bits)	Days	Cost	Bits/cost
Long-read WGS	0.582	35	4.00	0.15
Parental sequencing	0.350	21	1.00	0.35
RNA-seq (fibroblast)	0.288	28	2.00	0.14
Checkpoint functional assay	0.086	60	6.00	0.01

Two non-obvious results. **Parental sequencing is less decisive than intuition suggests**: it separates the proposed genotype from a cis configuration sharply, but separates it only weakly from hypotheses in which the missense is incidental, because under those its phase is close to a coin flip. And **the checkpoint functional assay has near-zero diagnostic value** — a defective checkpoint is expected under every hypothesis in contention — while being the highest-value class of experiment in the mechanistic problem. An experiment's worth is defined against a specified uncertainty, not in the abstract.

6.2 Mosaic aneuploidy burden from the call set

The selectivity argument in Section 5.2 turns on how aneuploid this patient's constitutional tissue actually is. That quantity was estimated from the provided VCF rather than assumed.

Method. At a heterozygous SNP in a purely diploid sample the alternate allele fraction is 0.5. If a fraction f of cells carry a trisomy, het sites split into branches at $(1+f)/(2+f)$ and $1/(2+f)$; for monosomy, $(1-f)/(2-f)$ and $1/(2-f)$. In both cases the absolute deviation from 0.5 is $f/(2(2\pm f)) \approx f/4$ for small f. Relative read depth provides a signed but noisier second signal. High-confidence biallelic het SNVs were retained (PASS, DP 15-100, GQ $\geq$ 30) and aggregated per chromosome.

Result. Sixteen chromosomes cluster tightly at mean absolute BAF deviation 0.0664-0.0691, consistent with pure binomial sampling noise at the observed depth.

Against that baseline, chr20 (0.0885), chr22 (0.0852), chr21 (0.0803) and chr17 (0.0770) are outliers, implying nominal mosaic fractions of 2-8%.

Interpretation, stated conservatively. The ranking — 20, 22, 21, 17, 9, 19, 16 — correlates with GC-rich chromosomes, the acrocentrics carrying rDNA arrays, and chr9 with its pericentromeric heterochromatin. That pattern is characteristic of mappability and alignment artefact rather than biology; genuine mosaic gains would not preferentially select the hardest-to-align chromosomes. The depth arm of the analysis is additionally unreliable at this resolution, because median depth is an integer and the resulting relative-depth granularity (4.5% in mosaic fraction) exceeds the effect size being measured.

Conclusion. Suggestive of low-level mosaicism, not established. The magnitude is biologically plausible — a published *BUB1B* MVA case reported 12% mosaicism on karyotype — but a validated mosaic chromosomal alteration caller with GC correction and within-chromosome segmentation would be required to separate signal from artefact. A null or ambiguous result is also expected to understate the true burden, since the sample is blood and proliferating haematopoietic lineages select against aneuploid cells.

This is a screen, not a diagnostic, and it is reported as one.

7. Clinical constraints

7.1 Immediately actionable

Genotoxic therapy. Because this is a chromosomal instability syndrome, standard genotoxic chemotherapy and radiotherapy for the rhabdomyosarcoma carry elevated risk of inducing secondary malignancy. This constrains how the existing tumour is treated and is independent of any speculative molecule discussed here.

7.2 Paediatric pharmacology, specific to this patient

- **Renal.** Nephrocalcinosis from birth. Nephrotoxic agents — aminoglycosides above all — are relatively contraindicated; renally cleared drugs may require dose adjustment.
- **Growth.** Severe IUGR, short stature, failure to thrive. Weight-based dosing on a child well below centile; agents suppressing appetite or growth are poorly tolerated.
- **Chronic versus acute exposure.** A rebalancing strategy would be lifelong. Toxicity tolerable across six cycles of cancer therapy is unacceptable across two decades.
- **On-target toxicity of the proposed class.** Aurora B is essential in all dividing tissue. Toxicity is therefore on-target and cannot be engineered away with a more selective compound. Aurora kinase inhibitors at oncology doses cause severe neutropenia, which is a serious consideration in a child with cancer predisposition.
- **Developmental pharmacology.** Hepatic enzyme maturation, body composition and renal clearance all differ from adults; adult PK does not transfer.

7.3 Standard of care

MVA management centres on intensive tumour surveillance — imaging protocols for Wilms tumour and rhabdomyosarcoma — because early detection currently dominates any available therapeutic option. Haematopoietic stem cell transplantation has been reported in an MVA1 patient for marrow failure. Any proposal in this document is adjunctive to, not a replacement for, that surveillance.

8. Limitations

Stated at length because this is where the argument should be attacked.

On the proposed thesis.

- The published rescue was **acute chemical inhibition in cultured cells**, not chronic therapy in an organism.
- The MVA patient lines in that work carried **their own mutations**, not p.Asn1002Lys. Whether the rescue transfers to this allele is untested and is the pivotal unknown.
- Whether p.Asn1002Lys is **hypomorphic or dominant-negative** is unresolved; the therapeutic logic differs between them.
- **No compound is nominated with PK modelling** against a measured target concentration, because the required degree of target engagement is unquantified.
- The **therapeutic window for chronic paediatric dosing is uncharacterised**. A U-shaped curve implies the window may be narrow, and the wrong side of it is actively harmful.

On the evidence base.

- Mechanistic reasoning relies on mouse, yeast and immortalised cell-line models; cross-species and cross-context transfer is assumed, not demonstrated.
- Several cited dependencies derive from cancer cell lines, which are poor proxies for constitutional tissue.
- No patient-derived cells were available, so no direct measurement of selectivity, binding or dose-response was possible.

On the supporting analyses.

- VoI priors are **elicited, not calibrated**; conclusions are conditional on them.
- VoI outcomes are **discretised** into three levels, which discards information and makes the result sensitive to bin placement.
- Experiments are **scored independently**; sequential and combinatorial designs are not evaluated and the greedy ranking is not guaranteed optimal.
- **Hypothesis closure is assumed**; no probability mass is reserved for an unanticipated mechanism.
- The VoI framework has **no external validation**. The appropriate benchmark is the CAGI6 Rare Genomes Project set (Stenton et al. 2024), on which the nominated experiment could be checked blind against the one that in fact resolved each case. That evaluation was not performed.
- The mosaic screen is **chromosome-level only** and will miss segmental and arm-level events entirely; its findings are confounded with mappability as described in Section 6.2.

9. Reproducibility

The repository contains the analysis code: Ensembl coordinate lookup, VCF region scan, genotype parsing, VEP annotation and candidate ranking for Track 1; the value-of-information implementation; and the mosaic aneuploidy burden screen. External services used are the Ensembl REST lookup and VEP endpoints.

No patient sequence data or genotype-scale derivative is included, and none will be. Reported findings are limited to the named candidate variants and chromosome-level aggregate statistics.

10. References

1. Hanks S, et al. Constitutional aneuploidy and cancer predisposition caused by biallelic mutations in *BUB1B*. *Nature Genetics*, 2004.
2. BUBR1 recruits PP2A via the B56 family of targeting subunits to promote chromosome congression (PMID 23789096), and Integration of kinase and phosphatase activities by BUBR1 ensures formation of stable kinetochore-microtubule attachments, *Developmental Cell*, 2012. **Central references for the rescue result in MVA patient-derived cells.**
3. BUBR1 pseudokinase domain promotes kinetochore PP2A-B56 recruitment, spindle checkpoint silencing, and chromosome alignment. *Cell Reports*, 2020.
4. Saurin AT. Kinase and phosphatase cross-talk at the kinetochore. *Frontiers in Cell and Developmental Biology*, 2018.
5. Baker DJ, et al. BubR1 insufficiency causes early onset of ageing-associated phenotypes and infertility in mice. *Nature Genetics*, 2004.
6. Baker DJ, et al. Increased expression of BubR1 protects against aneuploidy and cancer and extends healthy lifespan. *Nature Cell Biology*, 2013.
7. BubR1 alterations that reinforce mitotic surveillance act against aneuploidy and cancer. *eLife*, 2016; 5:e16620.
8. Overlack K, et al. A molecular basis for the differential roles of Bub1 and BubR1 in the spindle assembly checkpoint. *eLife*, 2015; 4:e05269.
9. Chen G, et al. Hsp90 stress potentiates rapid cellular adaptation through induction of aneuploidy. *Nature*, 2012.
10. Oromendia AB, Amon A. Aneuploidy: implications for protein homeostasis and disease. *Disease Models & Mechanisms*, 2014.
11. Sterling NA, et al. P53-independent pathogenic mechanisms contribute to BubR1 microcephaly. *Frontiers in Cell and Developmental Biology*, 2023.
12. Guney E, et al. Network-based in silico drug efficacy screening. *Nature Communications*, 2016.
13. Stenton SL, et al. Critical assessment of variant prioritization methods for rare disease diagnosis within the Rare Genomes Project. *Human Genomics*, 2024.
14. Mosaic variegated aneuploidy in development, ageing and cancer. *Nature Reviews Genetics*, 2024.

15. Robinson PN, et al. Interpretable clinical genomics with a likelihood ratio paradigm (LIRICAL). *American Journal of Human Genetics*, 2020.
16. Tavtigian SV, et al. Modeling the ACMG/AMP variant classification guidelines as a Bayesian classification framework. *Genetics in Medicine*, 2018.

Acknowledgements

This work was made possible through the Hackathon, organized by Sage Bionetworks in partnership with the MVA Society, Hugging Face, and BEACON (The Benchmarking, Evaluation, and Assessment Consortium for Science), with prize sponsorship from AWS and Anthropic. I am deeply grateful to the child and their family who generously contributed their data and their story to advance research into this rare disease.